Areeb Ahmad Chaudhry

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Google Scholar | GitHub | LinkedIn | Hugging Face

Research Interests

Multimodal and vision-language models for 2D and 3D medical imaging; anatomy- and knowledge-grounded generation; parameter-efficient adaptation of foundation models; reliable retrieval-augmented and agentic LLM systems and their evaluation.

Education

National University of Sciences and Technology (NUST), SEECS *Nov 2021 – Jun 2025*
Bachelor of Software Engineering Islamabad, Pakistan

- **Final Year Project:** Radixpert, an anatomy-guided vision-language model for radiology report generation. Carried out in the MachVis Lab; graded A in both project semesters and published at IEEE (see Publications).
- **Relevant coursework:** Large Language Models; Linear Algebra & ODEs; Probability & Statistics; Numerical Methods; Data Structures & Algorithms; Cloud Computing.

Publications

[1] M. A. R. Farooqi, **A. A. Chaudhry**, M. Y. Ghaffar, N. Perwaiz, and H. M. Kiani, "Radixpert: A Multimodal Approach to Medical Imaging," IEEE, 2025. [IEEE Xplore]

[2] M. Javed, **A. A. Chaudhry**, N. Perwaiz, U. Athar, A. Salahudeen, and M. M. Fraz, "FeLoRA16-SPP: Parameter-Efficient Fine-Tuning of 3D Multimodal LLMs for Radiology Report Generation and Visual Question Answering," MICCAI 2025 FLARE Challenge (Task 5) paper, 2025. [OpenReview]

Research Experience

MachVis Lab, SEECS, NUST *2024 – Sep 2025*
Research Assistant (Jun – Sep 2025); undergraduate researcher during final-year project. Islamabad, Pakistan
Worked with Dr. Nazia Perwaiz.

FeLoRA16-SPP: 3D Multimodal LLM for CT Report Generation and VQA *Jun 2025 – Sep 2025*

- Adapted the M3D-LaMed 3D multimodal LLM for radiology report generation and visual question answering on 3D CT scans (FLARE 2025, Task 5), combining LoRA fine-tuning with a spatial pooling projector and a frozen vision encoder; about 8% of 4.07B parameters trainable.
- Reached an average GREEN score of 0.431 across 18 organ systems, up to 12% above PHI3 and 4% above Med3DVLM, with top scores on 9 organs; accuracy on global VQA questions stayed below the baselines.

Radixpert: Anatomy-Guided VLM for Radiology (Final Year Project) *Sep 2024 – Jun 2025*

- Co-developed a dual-branch model that fuses frozen BioViL image features with anatomical segmentation masks from a frozen U-Net (via a 1×1 convolution and MLP projector) to condition MedGemma-4B for report generation.
- Adapted the language model with LoRA, training about 1.8% of parameters ($\sim$126M), on a 16,000-study subset of PadChest with Spanish-language reports.
- Reached RadCliQ 0.804, comparable to the strongest baseline (Reg2RG, 0.810); ablations isolated each component (0.682 base $\rightarrow$ 0.755 with LoRA $\rightarrow$ 0.804 with the anatomy branch), and explicit anatomical grounding reduced laterality errors.

Industry Experience

GoSaaS, AI Engineer | Remote *Jun 2025 – Present*

- Built computer vision pipelines (OpenCV, fine-tuned YOLO) to extract data from scanned invoices and industrial P&ID diagrams, and image-based semantic search matching part images to CAD drawings.
- Enabled plain-language querying of enterprise Oracle databases with knowledge-graph agents, fine-tuned NL2SQL LLMs and hybrid (semantic + keyword) OpenSearch/Elasticsearch retrieval.
- Shipped 8+ production ERP/CRM agents with built-in evaluation and monitoring on Oracle AI Agent Studio, published on the Oracle Marketplace.

Heartstamp, Founding AI Engineer, GTM | Las Vegas, USA (Remote) *Sep 2025 – Jul 2026*

- Architected the product's core multi-agent LangGraph system (support, design and creative agents) with checkpointed memory, structured outputs and provider-agnostic model routing.

- Built the evaluation and monitoring pipeline (DeepEval, LangSmith, OpenTelemetry) and hybrid semantic search (pgvector HNSW, full-text, RRF fusion, re-ranking) to catch regressions before release.

Codeacme, AI Engineer, Contract | Islamabad, Pakistan (Remote) *Mar 2025 – Sep 2025*
- Fine-tuned diffusion models with LoRA/PEFT and built end-to-end MLOps from training to serverless deployment, raising project delivery rate by 20%.

Pam AI, AI Support Engineer, Contract | Virginia, USA (Remote) *Dec 2024 – May 2025*
- Reduced hallucinations in a production voice-AI system with agentic, multi-tool pipelines, cutting responses that needed human escalation.

Tranzbox, Data Scientist, Quant | Melbourne, Australia (Remote) *Jun 2023 – Sep 2024*
- Benchmarked classical ML (SVM, tree ensembles) and reinforcement learning (A3C) for market prediction on large-scale stock data; built Airflow ETL and CI/CD on AWS; led a team of 4 engineers.

Selected Projects

- **NUST BankLlama:** LLaMA 3.2-3B fine-tuned with QLoRA for banking question answering on a self-hosted model, with Qdrant + BM25 hybrid retrieval and guardrails for PII detection and jailbreak prevention.
- **Scanned-document RAG:** Google Vision OCR and LlamaIndex pipeline with fine-tuned BGE-small embeddings (+4.3% retrieval) and hybrid BM25 + dense retrieval.
- **Research-paper RAG:** arXiv ingestion, semantic chunking, ensemble retrieval and Cohere re-ranking, tuned with RAGAS evaluation.

Technical Skills

Models & training:	LoRA/QLoRA/PEFT, SFT, vision-language and 3D multimodal LLMs (MedGemma, BioViL, M3D-LaMed), YOLO fine-tuning, OpenCV, diffusion fine-tuning
LLM systems:	LangGraph, LangChain, LlamaIndex, multi-agent systems, RAG, RAGAS, DeepEval, LangSmith
Retrieval & data:	pgvector, Qdrant, OpenSearch, Elasticsearch, BM25, Cohere Rerank, PostgreSQL, Apache Airflow
Engineering:	Python, FastAPI, Docker, AWS (Lambda, S3, EC2, MWAA), GitHub Actions, Celery, Redis
Writing & open source:	Technical articles on Medium; open models and datasets on Hugging Face